

MediScript-E — Presentation Slides

Software Engineering | SDLC + IEEE | USTC

✓ 22 Slides — All images injected, no orphan placeholders

SLIDE 1 — Title

✓ IMAGE: [Landing_page_1.png](#) — Right side

MediScript-E

A Secure Digital Healthcare Platform

Software Engineering Course Project

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SLIDE 2 — Abstract (IEEE)

✓ IMAGE: [Patient-Dashboard-overview-1.png](#) — Right side

What is MediScript-E?

- Full-stack digital healthcare platform
- Connects Patients, Doctors, and Administrators
- Built following complete SDLC methodology

Core Purpose:

- Digitize patient-doctor workflow end-to-end
- Secure medical data in the cloud
- Automate medicine reminders

Built with: Next.js · TypeScript · PostgreSQL · Vercel

SLIDE 3 — Outline (IEEE)

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#	Section	Phase
1	Abstract	IEEE
2	Related Work	IEEE
3	Problem Statement	SDLC Phase 1
4	Proposed Solution	SDLC Phase 1
5	Requirements Analysis	SDLC Phase 1
6	Use Cases & Actors	SDLC Phase 1
7	System Architecture	SDLC Phase 2
8	Database Design	SDLC Phase 2
9	Tech Stack	SDLC Phase 3
10	Auth & Security	SDLC Phase 3
11	Core Features	SDLC Phase 3
12	Testing	SDLC Phase 4
13	Results & Deployment	SDLC Phase 5
14	Challenges & Conclusion	IEEE

SLIDE 4 — Related Work (IEEE)

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Existing System	What It Does	What's Missing
Practo (India)	Doctor discovery & appointment booking	No prescription management, no reminders, no vault
Zocdoc (USA)	Online appointment scheduling	No e-prescriptions, no 2FA, no AI chatbot
HealthTap	AI symptom checker + doctor chat	No appointment booking, no prescription PDF, no RBAC
Manual / Paper-based	Traditional in-person healthcare	No digitization — prescriptions lost, records scattered

Gap Identified:

- No existing free platform combines: appointment booking + e-prescription + reminders + vault + AI chatbot
- Most platforms lack RBAC and 2FA
- MediScript-E fills this gap — open, secure, full-stack, production-deployed

SLIDE 5 — Problem Statement (SDLC Phase 1)

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Pain Point	Reality
Prescriptions	Lost, damaged, inaccessible
Appointments	Phone calls, physical visits
Medical Records	Scattered across hospitals
Medicine Schedule	No reminders, often forgotten
Data Security	No role-based access

"Healthcare data is fragmented, insecure, and inaccessible when needed most."

SLIDE 6 — Proposed Solution (SDLC Phase 1)

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One platform. Three roles. Complete workflow.

Role	Key Capability
 Patient	Book appointments, view prescriptions, set reminders, upload records
 Doctor	Manage appointments, issue digital prescriptions
 Admin	Monitor platform, manage users

25+ APIs · 9 DB Models · 3 Roles · Live on Vercel

SLIDE 7 — Requirements Analysis (SDLC Phase 1)

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Functional Requirements

- FR1: Register & login with email verification
- FR2: Book appointments with available doctors
- FR3: View & download prescriptions as PDF
- FR4: Set medicine reminders — automated email alerts
- FR5: Upload & manage medical records (cloud)
- FR6: Doctor manages appointments & prescriptions
- FR7: Admin monitors users & platform stats

Non-Functional Requirements

NFR	Requirement
Security	2FA, RBAC, bcrypt, JWT

NFR	Requirement
Performance	Serverless, connection pooling
Scalability	Vercel + Supabase managed infra
Usability	Responsive, mobile-friendly UI
Reliability	Token expiry, cascade delete
Maintainability	TypeScript, Prisma ORM

SLIDE 8 — Use Cases & Actors (SDLC Phase 1)

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Actors: Patient · Doctor · Admin · Cron System · OAuth Provider

Key Use Cases:

- UC1: Register → Verify Email → Login
- UC2: Patient books appointment
- UC3: Doctor confirms & issues prescription
- UC4: Patient downloads prescription PDF
- UC5: Patient sets reminder → receives email
- UC6: Admin manages users & monitors platform

Appointment Flow: [PENDING](#) → [CONFIRMED](#) → [COMPLETED](#) or [CANCELLED](#)

SLIDE 9 — System Architecture (SDLC Phase 2)

✓ IMAGE: [Landing-page-3.png](#) — Right side

Three-Tier Architecture

- Presentation Tier: Next.js App Router · React 19 · TypeScript
- Application Tier: Next.js Serverless API Routes · NextAuth · Prisma ORM
- Data Tier: PostgreSQL (Supabase) · Supabase Storage

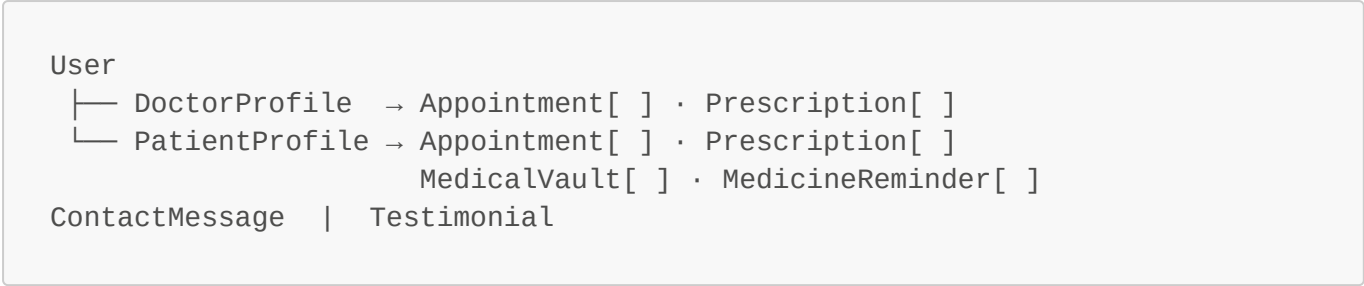
External Services:

- Groq API — AI Chatbot (Llama 3.1)
- Nodemailer — Email (OTP, Verification, Reminders)
- OAuth — Google & GitHub
- GitHub Actions — Cron job (medicine reminders)

SLIDE 10 — Database Design (SDLC Phase 2)

✓ IMAGE: [Landing-page-4.png](#) — Right side

9 Models · 3 Roles · Cascade Delete



Key Decisions:

- Soft archive on prescriptions (**archivedByDoctor**)
- OTP stored temporarily — cleared after verification
- Cascade delete — no orphan records

SLIDE 11 — Tech Stack (SDLC Phase 3)

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Layer	Technology	Purpose
Framework	Next.js 16 (App Router)	Full-stack web framework
Language	TypeScript 5	Type safety
Styling	Tailwind CSS 4 + Framer Motion	UI & animations
Auth	NextAuth.js 4	JWT sessions + OAuth
ORM	Prisma 7	Type-safe DB queries
Database	PostgreSQL via Supabase	Managed relational DB
Storage	Supabase Storage	Medical file uploads
Email	Nodemailer (Gmail SMTP)	Verification, OTP, reminders
AI	Groq SDK — Llama 3.1 8B	MediBot chatbot
Deployment	Vercel	Serverless hosting + CI/CD

SLIDE 12 — Authentication & Security (SDLC Phase 3)

✔ IMAGE (top): **2FA-OTP-screen.png** ✔ IMAGE (bottom): **Landing-page-05.png**

1. **Email Verification** — crypto token · 24h expiry
2. **Password Hashing** — bcryptjs · 10 salt rounds
3. **Two-Factor Auth (2FA)** — email OTP · 10 min expiry
4. **OAuth** — Google & GitHub · auto-verified
5. **JWT Sessions** — 30-day expiry · role in token
6. **RBAC** — every API route checks role + ownership

SLIDE 13 — Patient Features (SDLC Phase 3)

 IMAGE (card 1): Patient-Dashboard-overview-1.png

 IMAGE (card 2): Landing-page-6.png

 IMAGE (card 3): Landing-page-7.png

 IMAGE (card 4): Landing-page-8.png

1. **Appointment Booking** — Select doctor → pick date & time → confirm
2. **E-Prescription** — View diagnosis + medications → download as PDF
3. **Medicine Reminders** — Set schedule → automated email alerts daily
4. **Medical Vault** — Upload reports → stored securely in cloud

SLIDE 14 — Doctor & Admin Features (SDLC Phase 3)

 IMAGE (left): Doctor-Dashboard-overview-1.png

 IMAGE (right): Admin-Dashboard-overview-1.png

Doctor

- Confirm / cancel / complete appointments
- Issue prescriptions → appointment auto-completed
- Edit, archive, delete prescriptions

Admin

- Real-time platform statistics
- Delete users (cascade — all data removed)
- Monitor all appointments & contact messages

SLIDE 15 — MediBot & Search (SDLC Phase 3)

 IMAGE: MediBot.png — Left bottom

MediBot — AI Chatbot

- Powered by Groq (Llama 3.1 8B Instant)
- Available globally on all pages
- Responds in Bangla if user writes in Bangla

Global Search

- Trigger: navbar icon or **Ctrl+K**
- Role-aware results · Debounced 300ms

SLIDE 16 — Testing (SDLC Phase 4)

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Test Area	Result
Auth flows (register, login, 2FA, OAuth)	 Pass
Appointment lifecycle	 Pass

Test Area	Result
Prescription CRUD	✓ Pass
Medicine reminder + email delivery	✓ Pass
Medical vault upload/delete	✓ Pass
Security & authorization checks	✓ Pass
Edge cases (past date, duplicate email, expired OTP)	✓ Pass

SLIDE 17 — Results & Evaluation (SDLC Phase 5)

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Feature	Status
Email Verification (24h token)	✓ Delivered
Two-Factor Authentication (2FA)	✓ Delivered
Google & GitHub OAuth	✓ Delivered
Appointment Booking & Management	✓ Delivered
Digital Prescriptions + PDF Download	✓ Delivered
Automated Medicine Reminders (email)	✓ Delivered
Medical Vault (cloud storage)	✓ Delivered
AI Chatbot — MediBot (Groq)	✓ Delivered
Global Search (role-aware)	✓ Delivered
Admin Dashboard & User Management	✓ Delivered
Responsive UI (mobile + desktop)	✓ Delivered
Production Deployment — Live on Vercel	✓ Delivered

All SDLC phases completed. All requirements met.

SLIDE 18 — Deployment (SDLC Phase 5)

✓ IMAGE: `vercel-deployment-dashboard.png` — Right side

Push to GitHub → Vercel auto-build → Live on Vercel

- Database: Supabase PostgreSQL + PgBouncer pooling
- Cron: GitHub Actions — every 5 min → reminder emails
- Env vars: 10 variables configured on Vercel

 <https://mediscript-e.vercel.app>

SLIDE 19 — Challenges & Solutions (IEEE)

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Challenge	Solution
2FA re-auth after OTP	Magic token + DB OTP null check
Supabase SSL on Vercel	<code>sslmode=no-verify</code> + PgBouncer URL
OAuth users — no password	Check empty password before update
Reminder time zone mismatch	Convert to UTC+6 (Bangladesh) server-side
Prisma connection in serverless	Singleton via <code>globalThis</code>
Cascade delete order	Manual delete in dependency order
Stale JWT after profile update	Force <code>window.location.reload()</code>

SLIDE 20 — Future Enhancements (IEEE)

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Short Term — Real-time notifications · Doctor availability calendar · Rating system

Medium Term — Video consultation (WebRTC) · Prescription QR code · Mobile app

Long Term — AI symptom checker · Payment gateway · HIPAA / GDPR compliance

SLIDE 21 — Conclusion (IEEE)

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SDLC Phases Completed: Requirements → Design → Implementation → Testing → Deployment ✓

Delivered: Secure, production-ready healthcare platform 3 roles · 25+ APIs · 9 DB models · Live on Vercel

"Structured Software Engineering methodology turns ideas into maintainable, scalable, real-world applications."

SLIDE 22 — Q&A

🎥 [MANUAL: Insert demo video in PowerPoint]

Thank You

MediScript-E — *"Your Health, Digitally Managed."*